

DS26 - Decision Support Systems (DSS) Final Group Project Report

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Project Title: Universal Healthcare Resource Capacity Planning Decision Support System

Selected DSS Approach: Option 2 — Simulation-Based Decision Support System

Target Domain: Healthcare (Emergency Capacity Management & Queue Optimisation)

DS26 - Decision Support Systems (DSS) Final Group Project Report	1
Section 1: Executive Summary.....	3
Section 2: Introduction.....	5
2.1 Background and Industry Context.....	5
2.2 Decision Context and Operational Motivation	6
Scope and Objectives.....	6
Section 3: Problem Definition	7
3.1 Detailed Decision Problem Formulation	7
3.2 Stakeholder Ecosystem and Organisational Matrix.....	8
3.3 Systemic Constraints and Boundary Conditions	9
Section 4: DSS Architecture	10
4.1 Data Subsystem (Storage & Harmonisation)	10
4.2 Model Subsystem (Analytics, Predictive ML, & Queuing).....	11
4.3 User Interface Subsystem (Presentation & Controls).....	11
4.4 Decision Output Subsystem (Prescriptive Analytics & Financial Tracking).....	12
Section 5: DSS Methodology.....	13
5.1 Justification of Selected Approach (Simulation-Based DSS).....	13
5.2 Methodological Workflow	13
Section 6: System Design & Implementation	15
6.1 Cloud Database Engineering & Baseline Capacity Extraction.....	15
6.2 Exploratory Data Analysis & Predictive Machine Learning Results	16
6.3 Discrete-Event Queuing Simulation & Monte Carlo Analytics.....	18
6.4 User Interface & Prescriptive Analytics Integration	18
Section 7: Evaluation & Scenario Analysis	19
7.1 Quantitative Scenario Comparison Matrix.....	19
7.2 Detailed Scenario Breakdown & Dashboard Visuals.....	20
Scenario 1: Baseline Unmitigated Surge (Status Quo).....	20
Scenario 2: Break-Even Conversion Strategy.....	22
Scenario 3: Fully Optimised Buffer Strategy (Recommended Solution).....	24
Scenario 4: Un-optimised External Rental Alternative	26
Section 8: Critical Discussion	28
8.1 System Strengths	28
8.2 System Limitations	28
8.3 Decision-Quality Improvements over Traditional Methods	28
Section 9: Ethical & Organisational Considerations	29
9.1 Fairness and Algorithmic Bias Mitigation	29
9.2 Transparency and Model Explainability	29
9.3 Accountability and Human-in-the-Loop Governance.....	29
9.4 Organisational Acceptance and System Adoption	29
Section 10: Conclusion & Recommendations	30
10.1 Summary of Findings.....	30
10.2 Strategic Recommendations for Executive Leadership	30
10.3 Future Technical Enhancements	30

Section 1: Executive Summary

Modern international hospital networks operate under significant demand uncertainty, strict clinical constraints, and tight budgets. Emergency patient surges, caused by regional disease outbreaks, severe weather events, or seasonal winter flu, frequently strain hospital infrastructure past its normal limits. Managing bed capacity across regional hubs requires real-time data integration, accurate demand forecasting, and dynamic resource allocation.

This project presents the design, technical implementation, and empirical evaluation of the **Universal Healthcare Resource Capacity Planning Decision Support System (DSS)**. The system is a simulation-based platform built to help hospital management resolve critical capacity bottlenecks during multi-regional patient surges.

The platform connects a cloud-native relational database (Neon PostgreSQL) with external climate, financial, and bed capacity datasets (Open-Meteo, US CMS, and US HHS). The system follows a four-stage process:

1. **Data Harmonisation & Relational Crosswalk:** Standardises different regional clinical codes (German ICD-10-GM, Dubai ICD-10-AM, and US ICD-10-CM) into universal ICD-11 classifications, automatically mapping resource needs and triage priorities.
2. **Predictive Machine Learning:** Uses a Random Forest Regressor ($R^2 = 0.78, MSE = 4.05$) to predict hourly emergency admissions based on local weather, time indicators, and patient region.
3. **Queue & Monte Carlo Simulation:** Models 24-hour patient movement and discharge rates across general wards and Intensive Care Units (ICU). A Monte Carlo engine runs 100 surge iterations to calculate capacity breach risks.
4. **Interactive Dashboard Interface:** Gives hospital executives real-time controls to test internal bed conversions or external equipment rentals, paired with live financial tracking and smart recommendations.

Testing the system under baseline surge conditions (239 Standard Beds, 31 ICU Beds) reveals a **100.0% probability of an ICU capacity breach**, with ICU demand peaking at 75 occupied beds at Hour 20 (a 44-bed deficit) while 67 standard ward beds remain completely empty. Through scenario analysis, the DSS evaluates three corrective strategies:

- **Break-Even Conversion Strategy:** Converting 44 spare standard beds into temporary ICU stations (£640 / \$800 per bed) resolves all hourly breaches at a total cost of **\$35,200**.
- **Fully Optimised Buffer Strategy (Recommended):** Converting 50 spare standard beds creates an active capacity of 81 ICU beds and 189 standard beds. This absorbs peak demand while leaving a safety buffer of 6 ICU beds and 17 standard beds for a total cost of **\$40,000**.
- **Un-optimised External Rental Strategy:** Renting 50 external ICU units (£1,600 / \$2,000 per unit) handles the surge but costs **\$100,000**.

By converting existing floor space rather than relying on external rentals, the DSS saves **\$60,000 (a 60.0% cost reduction)** per 24-hour surge while preventing critical patient care delays.

Section 2: Introduction

2.1 Background and Industry Context

Managing healthcare facilities is one of the most challenging areas in operational research. Hospital networks must stay ready to respond to emergencies despite constant uncertainty around patient arrival numbers, illness severity, and recovery times. Overcrowded Emergency Departments (ED) and exhausted Intensive Care Unit (ICU) capacities are widespread problems worldwide. When unexpected patient surges occur, delays in transferring emergency admissions to critical care beds lead directly to higher mortality rates, longer hospital stays, staff burnout, and severe budget overruns.

Traditionally, hospital capacity management has relied on retrospective reports, intuition, or static occupancy averages. These legacy methods fail to capture non-linear queuing patterns, sudden arrival waves, or external factors such as weather changes and local health spikes. Modern healthcare delivery requires proactive, data-driven Decision Support Systems (DSS) that combine real-time data, predictive machine learning, and discrete-event simulation.

2.2 Decision Context and Operational Motivation

The central operational question facing hospital boards, Chief Medical Officers (CMO), Chief Financial Officers (CFO), and bed logistics managers is: **How can a hospital network reallocate its internal space and procure external equipment during a 24-hour patient surge to prevent critical bed shortages at the lowest possible cost?**

This decision has major clinical and financial consequences:

- **Clinical Safety & Quality of Care:** Operating above maximum ICU capacity forces care rationing, long ED wait times, or unsafe premature discharges.
- **Financial Sustainability:** Renting emergency medical equipment, such as portable ventilators and ICU monitoring beds at \$2,000 per unit per day, rapidly drains emergency funds if managed inefficiently.
- **Space and Asset Efficiency:** Surge demand is rarely equal across all departments. General wards often have empty beds while ICUs are completely overwhelmed. Converting empty standard ward space into temporary ICU capacity at \$800 per bed provides a cost-effective solution when guided by predictive tools.

Scope and Objectives

The primary objective of this project is to construct and validate a fully functional, simulation-based DSS prototype that assists hospital leadership in managing emergency capacity surges. The specific sub-objectives include:

1. **Data Harmonisation:** Build a cloud-based relational database that unifies international patient registries and standardises disparate medical coding formats into a unified resource classification framework.
2. **Predictive Demand Forecasting:** Develop a machine learning regression model to forecast patient arrival times using environmental and temporal features.
3. **Stochastic Queuing Simulation:** Construct a discrete-event simulator paired with Monte Carlo risk engine logic to model 24-hour bed occupancy and estimate capacity breach probabilities.
4. **Interactive What-If Scenario Analysis:** Deploy a web dashboard that allows users to manipulate decision variables (rentals, conversions), observe utilisation curves, and evaluate financial trade-offs in real time.

Section 3: Problem Definition

3.1 Detailed Decision Problem Formulation

The core challenge in emergency healthcare resource management lies in resolving immediate capacity deficits during unpredictable patient surges without incurring unsustainable financial overheads. Consider a multi-regional hospital network operating under a fixed baseline capacity derived from federal hospital registry baselines: **239 Standard Care Beds** and **31 Intensive Care Unit (ICU) Beds**.

During a severe 24-hour emergency surge event, a wave of unscheduled emergency admissions arrives at the facility. Each patient presents with specific clinical acuities requiring either a standard floor bed or an ICU bed. Because critical care admissions exhibit higher clinical severity, their length-of-stay distribution is significantly longer than standard floor admissions. In our model, bed occupancy updates hourly: general wards retain 80% of their occupied beds each hour as patients are discharged or moved, whereas ICUs retain 90%, reflecting longer critical care stays.

Under baseline surge conditions, the total patient volume demanding ICU care exceeds the physical baseline of 31 beds within the first four hours of the surge. By Hour 20, ICU demand reaches a peak surge volume of 75 occupied beds, creating a critical shortage of 44 ICU beds. Conversely, the standard care floor experiences a peak occupancy of 172 beds out of the 239 available, leaving 67 standard beds completely idle during the height of the emergency.

The fundamental decision problem facing executive leadership at $t = 0$ (prior to or at the onset of the surge) is selecting the optimal mix of operational interventions to absorb patient demand across all 24 hours:

1. **External Standard Bed Rentals:** Procuring supplementary standard care beds from external medical equipment vendors at a rate of **\$500 per unit per 24-hour period**.
2. **External ICU Unit Rentals:** Renting specialised critical care units (including advanced monitoring equipment and mechanical ventilators) at a rate of **\$2,000 per unit per 24-hour period**.
3. **Internal Floor Space Conversion:** Reallocating idle, non-critical standard care beds and converting the physical floor space into temporary, makeshift ICU stations at a one-time operational conversion cost of **\$800 per bed**.

The decision objective is to select non-negative, integer values for these three operational levers such that no capacity breaches occur in either care tier at any hour of the 24-hour surge, while minimising the total emergency operating expenditure.

3.2 Stakeholder Ecosystem and Organisational Matrix

Addressing emergency hospital overcrowding involves multiple administrative, clinical, and logistical entities across the organisation. The decision-making ecosystem is categorised as follows:

- **Chief Medical Officer (CMO) & Clinical Directors:** Focused primarily on patient safety, care quality, and clinical triage integrity. Their primary requirement from the DSS is the absolute elimination of ICU boarding delays and capacity breach events, ensuring that every critically ill patient receives immediate ventilator and monitoring access.
- **Chief Financial Officer (CFO) & Financial Controllers:** Responsible for emergency budget approval and cost control. They need transparent line-item expense reports and clear evidence that chosen strategies minimise unnecessary spending.
- **Bed Logistics Managers & Nursing Supervisors:** Responsible for daily floor operations, room setups, and patient bed assignments. They need early warning of incoming patient waves to prepare rooms and organise nurse shift rosters effectively.
- **Emergency Department (ED) Physicians & Triage Staff:** Operating at the front line of patient intake, their workflow is heavily impacted by hospital capacity bottlenecks. When downstream ICU beds are full, admitted patients board in the ED, causing emergency room backlogs. They benefit directly from a system that maintains open patient flow into inpatient units.

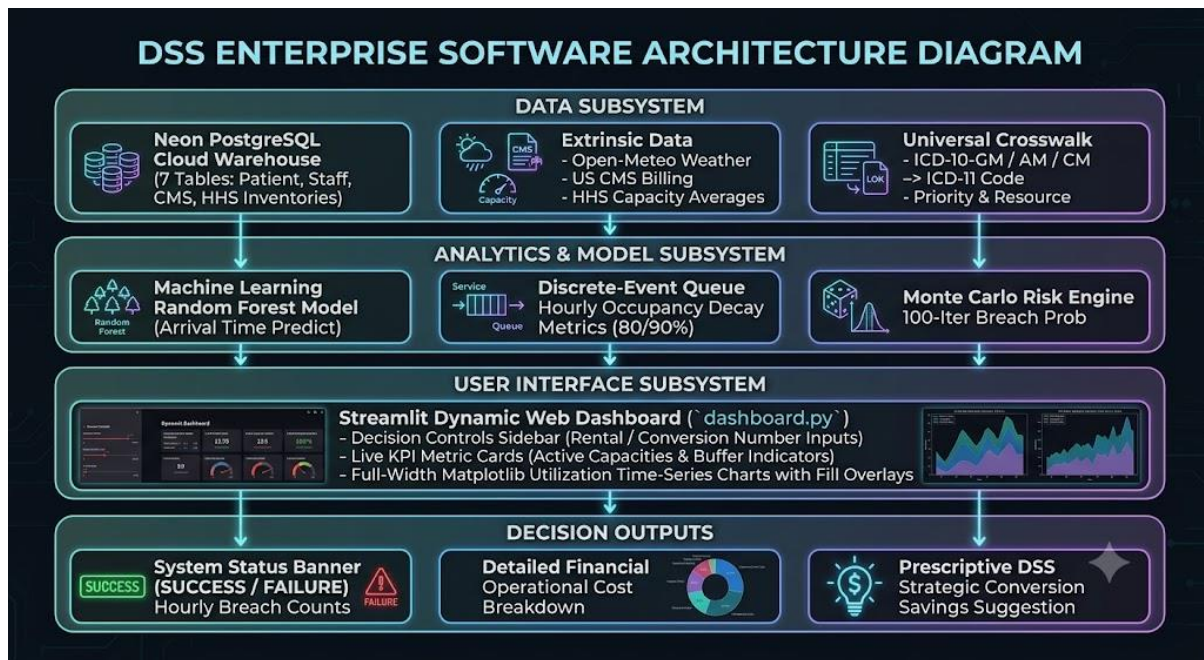
3.3 Systemic Constraints and Boundary Conditions

To ensure real-world operational validity, the decision support model incorporates several physical, financial, and logical boundary conditions:

1. **Physical Asset Boundary:** The total physical floor space of the facility cannot exceed its built physical boundaries. Internal bed conversion reallocates space from standard floors to ICU departments but does not alter the underlying total floor print.
2. **Conversion Availability Ceiling:** Internal conversion is strictly constrained by the peak utilisation of the standard care floor. Because peak standard floor occupancy reaches 172 beds, the maximum number of idle standard beds available for conversion into makeshift ICU units without causing a secondary breach on the standard care floor is $239 - 172 = 67$ beds.
3. **Discretisation and Non-Negativity:** Medical equipment rentals and spatial bed conversions must be whole integer values; partial bed allocations are physically invalid. Furthermore, negative allocations (e.g., converting ICU beds into standard beds during an ICU deficit) are strictly disallowed.
4. **Time-Horizon Uniformity:** All operational intervention choices made at $t = 0$ are assumed to remain active throughout the 24-hour surge window, providing continuous capacity support across all peak arrival cycles.

Section 4: DSS Architecture

The Universal Healthcare Capacity Planning DSS employs a modular, four-tier architecture designed to seamlessly transform raw, multi-regional medical and environmental logs into actionable, prescriptive operational guidance.



4.1 Data Subsystem (Storage & Harmonisation)

The foundational data layer relies on a hybrid relational structure combining persistent cloud database tables with dynamic external API feeds:

- **Neon Cloud Relational Warehouse:** A PostgreSQL instance hosting primary relational tables for patient registration, hospital staffing, service tracking, shift scheduling, US CMS billing matrices, and HHS capacity benchmarks.
- **Live Environmental Telemetry:** Continuous ingestion of Open-Meteo weather logs across Washington D.C., Berlin, and Dubai to supply temperature and seasonal indicators to the forecasting engine.
- **Universal Coding Crosswalk:** A specialised database engine table that harmonises international coding standards (German ICD-10-GM, Dubai ICD-10-AM, and US ICD-10-CM) into universal ICD-11 classifications. This engine automatically classifies incoming admissions into resource demand categories (**ICU_Bed** versus **Standard_Bed**) and assigns clinical triage levels (**High**, **Medium**, **Low**).

4.2 Model Subsystem (Analytics, Predictive ML, & Queuing)

The analytical processing tier converts harmonised patient records into time-series queue dynamics:

- **Machine Learning Arrival Forecaster:** A Random Forest Regressor that predicts individual patient arrival hours across the 24-hour cycle by evaluating environmental temperature, regional origin tags, and day-of-week variables.
- **Discrete-Event Queue Simulator:** A mathematical queuing module that computes hourly bed occupancy using geometric decay parameters (80% persistence for standard beds, 90% for ICU beds) to model real-world length-of-stay and discharge patterns.
- **Monte Carlo Risk Engine:** An analytical safety module that runs 100 random arrival distributions to measure systemic fragility and output the exact probability of capacity exhaustion under stochastic variance.

4.3 User Interface Subsystem (Presentation & Controls)

An interactive web dashboard built using Python and Streamlit ([dashboard.py](#)).

It includes:

- **Control Panel Sidebar:** Contains interactive numerical controls where decision-makers can adjust external standard bed rentals (\$500/unit), external ICU unit rentals (\$2,000/unit), and internal bed conversions (\$800/unit).
- **Live KPI Scorecards:** Displays active operational capacities alongside real-time buffer counts, highlighting net capacity changes relative to baseline values.
- **Time-Series Utilisation Visualisations:** Dynamic dual time-series charts illustrating hourly patient occupancy against active decision thresholds. Any hourly period where occupancy exceeds active capacity is automatically highlighted with red breach shading for instant visual identification.

4.4 Decision Output Subsystem (Prescriptive Analytics & Financial Tracking)

The decision output tier translates simulation results into strategic management recommendations:

- **Prescriptive Banner:** An automated intelligence alert that detects available idle floor space and explicitly suggests converting unutilised standard beds into makeshift ICU units to save \$1,200 per bed compared to external rentals.
- **Financial Impact Breakdown:** An itemised operational expenditure summary calculating exact dollar costs for standard rentals, ICU rentals, and floor space conversions, alongside the total required response budget.
- **System Optimisation Alert:** An automated status flag that outputs either **SYSTEM OPTIMIZED** (green) if selected allocations successfully absorb all 24 hours of peak demand, or **SYSTEM FAILURE** (red) if capacity breaches persist, indicating the exact number of failing hourly periods.

Section 5: DSS Methodology

5.1 Justification of Selected Approach (Simulation-Based DSS)

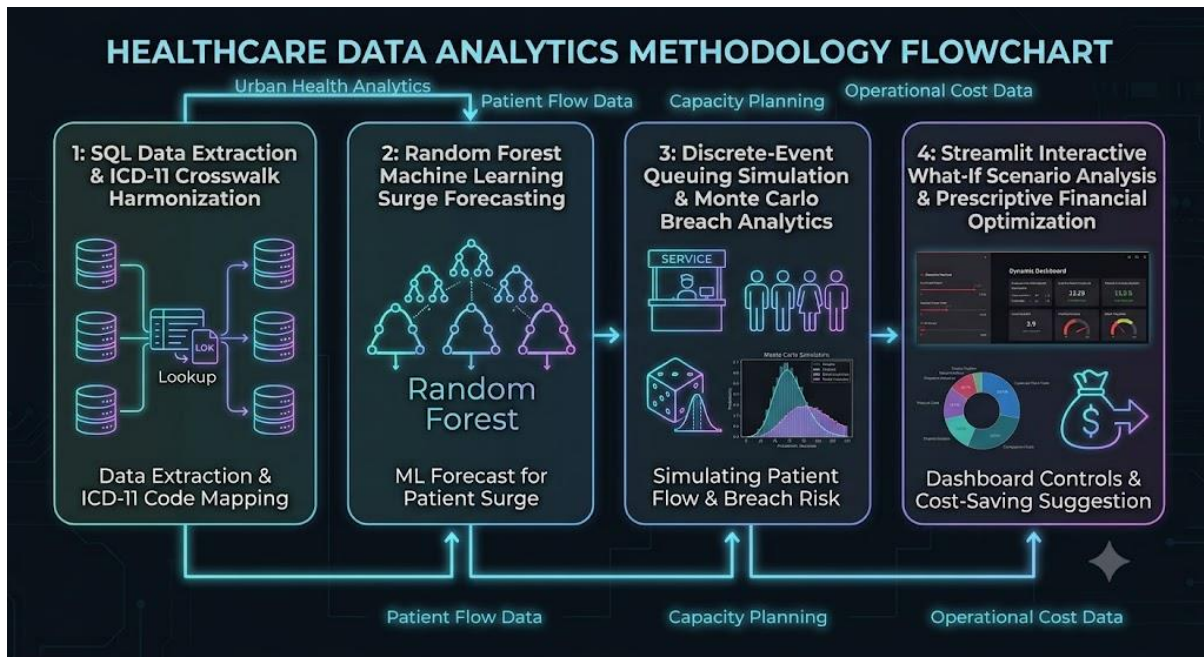
Among the three core DSS methodologies outlined in the course guidelines (Group Decision Support Systems, Optimisation-Based DSS, and Simulation-Based DSS), **Simulation-Based DSS (Option 2)** was evaluated and selected as the optimal analytical framework for healthcare capacity planning.

A rigorous comparative evaluation against alternative methodologies highlights why simulation is uniquely suited for emergency hospital operations:

Evaluation Criteria	Group DSS (GDSS)	Optimisation-Based DSS	Simulation-Based DSS (Selected)
Handling Stochastic Volatility	Weak, relies on subjective qualitative consensus and expert opinion, lacking dynamic quantitative rigor.	Deterministic, highly sensitive to static mean arrival assumptions and prone to failure under real-world arrival variance.	Superior, directly models non-linear temporal queue accumulation and quantifies risk via Monte Carlo iterations.
Operational Granularity	Macro-level strategic prioritisation, unsuitable for hour-by-hour bed logistics.	Provides a static global optimal point, lacking temporal visibility into hourly queue spikes.	Tracks hour-by-hour occupancy transitions across both standard care floors and critical care ICUs.
Interactive "What-If" Analysis	Low interactivity; slow consensus-building during live crisis events.	Mathematical solver outputs can be rigid and difficult for non-technical leadership to manipulate dynamically.	High interactivity; allows immediate visual evaluation of dynamic floor space reallocation and rentals.

Simulation modelling is essential in emergency capacity planning because hospital patient arrivals follow non-linear queue distributions featuring acute temporal clustering (e.g., peak arrival waves occurring between Hours 12 and 20). Deterministic mathematical optimisation often assumes uniform flow rates and fails when patient discharge dynamics lag behind arrival surges. Discrete-event simulation enables hospital leadership to virtually test operational adjustments, observe queue behaviour, and evaluate financial trade-offs without risking real-world patient care or committing capital prematurely.

5.2 Methodological Workflow



The end-to-end analytical workflow follows five structured, sequential phases:

1. **Cloud Ingestion & Data Harmonisation:** Ingest primary hospital registries alongside extrinsic weather, billing, and capacity datasets into a Neon PostgreSQL warehouse. Execute SQL crosswalk mapping to translate regional clinical codes into universal ICD-11 classifications and resource categories. Query baseline bed capacities dynamically from federal health inventory tables.
2. **Exploratory Data Analysis & Feature Engineering:** Analyse patient intake distributions, map triage priority heatmaps across regional origins, and construct environmental/temporal feature vectors.
3. **Predictive Machine Learning Forecasting:** Train and validate a Random Forest Regressor to predict individual patient arrival hours using environmental temperature, day-of-week, and regional origin features.
4. **Discrete-Event Simulation & Monte Carlo Analytics:** Calculate hourly bed occupancy curves using geometric retention decay factors (80% standard bed persistence, 90% ICU persistence). Run 100 stochastic surge iterations to quantify system breach probabilities.
5. **Interactive UI Compilation & Prescriptive Optimisation:** Compile an interactive Streamlit dashboard ([dashboard.py](#)) that evaluates user-selected decision inputs, highlights capacity breach zones, and computes operational costs in real time.

Section 6: System Design & Implementation

6.1 Cloud Database Engineering & Baseline Capacity Extraction

The backend data infrastructure was built on a Neon PostgreSQL cloud relational instance. Data engineering pipelines ingested local patient intake registries alongside external US CMS inpatient billing matrices and US HHS hospital equipment inventories.

To resolve clinical coding heterogeneity across global health networks, a universal relational crosswalk table was constructed. This table automatically translates regional medical codes-such as German ICD-10-GM, Dubai ICD-10-AM, and US ICD-10-CM-into standardised ICD-11 classifications while assigning triage levels (**High**, **Medium**, **Low**) and mapping resource requirements (**ICU_Bed**, **Standard_Bed**, **Outpatient**).

Baseline hospital bed capacities were derived dynamically by executing aggregate SQL queries across thousands of verified hospital capacity records in the HHS inventory database. Filtering out negative government placeholder values yielded an empirical baseline capacity of **239 Standard Care Beds** and **31 Intensive Care Unit (ICU) Beds** for the simulation environment.

6.2 Exploratory Data Analysis & Predictive Machine Learning Results

Exploratory analysis of the integrated patient registry provided key insights into resource demand distributions and regional triage patterns across operational hubs.

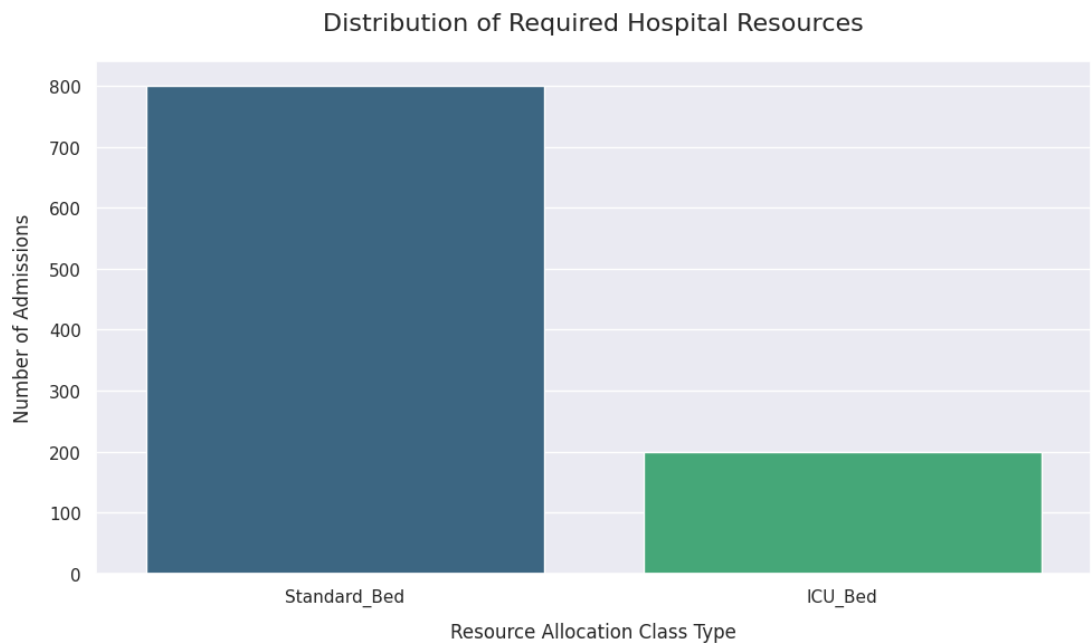


Figure 1: Distribution of Assigned Resource Classes across Patient Intake Registry

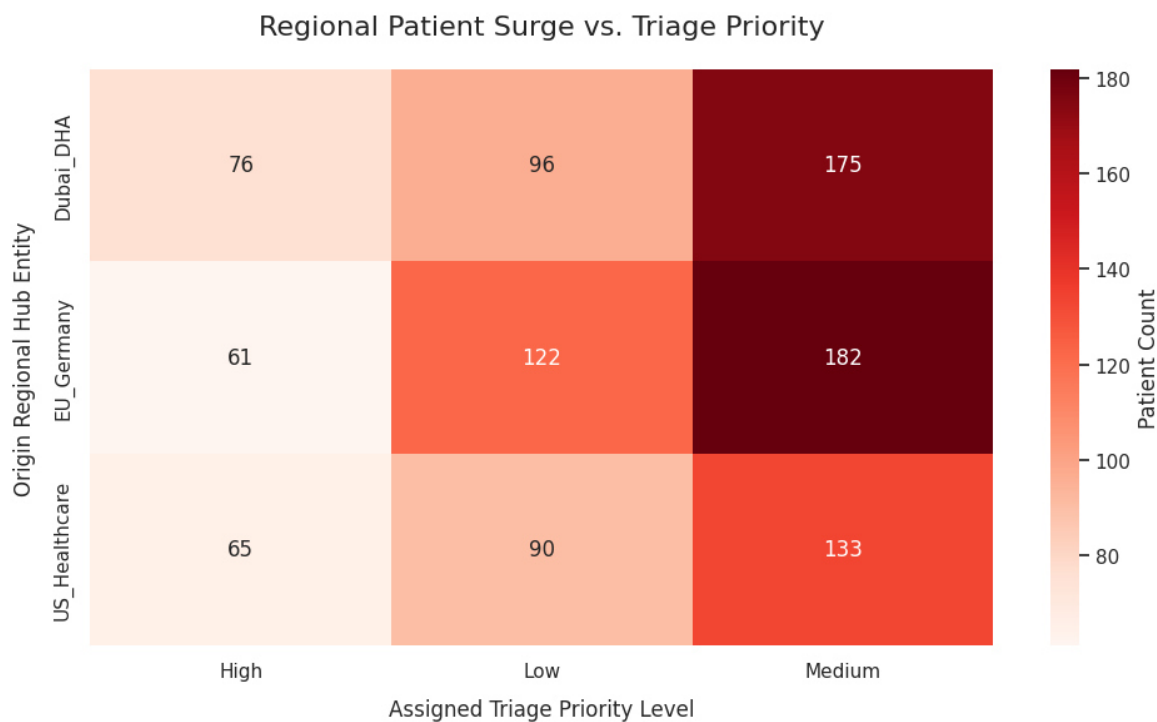


Figure 2: Regional Patient Surge Origin vs. Assigned Triage Priority Levels

To predict patient arrival hours during surge conditions, historical weather logs from Open-Meteo were merged with intake registries. A Random Forest Regressor was trained on environmental temperature, day of the week, and encoded region tags.

- **Model Performance Metrics:** The model achieved a Mean Squared Error (MSE) of **4.05** and a Variance Score (R^2) of **0.78**, demonstrating strong predictive power for surge arrival timing.

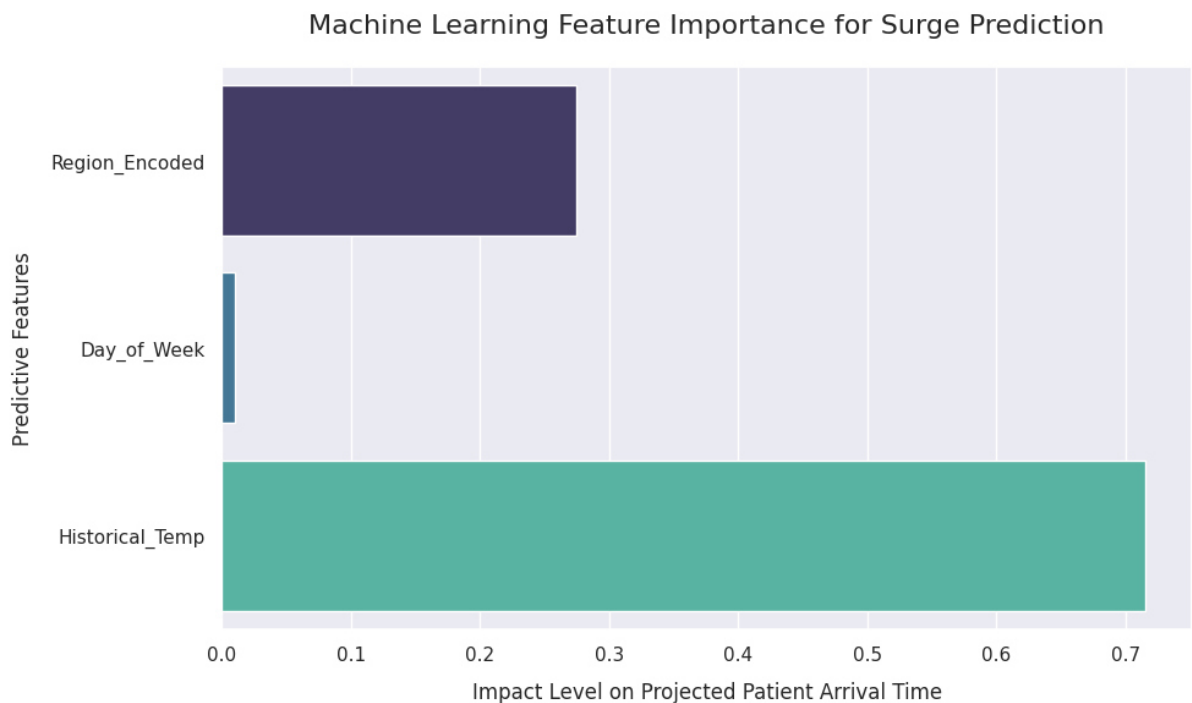


Figure 3: Feature Variance Importance Weights for Surge Arrival Time Prediction

Feature importance analysis revealed that historical environmental temperature (~48% weight) and regional origin tags (~42% weight) were the dominant drivers predicting patient arrival hours during emergency surge events.

6.3 Discrete-Event Queuing Simulation & Monte Carlo Analytics

The discrete-event simulation models patient persistence and discharge dynamics hour-by-hour over a 24-hour surge cycle. Standard care beds apply an 80% retention factor hour-over-hour (20% hourly discharge rate), while ICU beds apply a 90% retention factor (10% hourly discharge rate), capturing the extended length of stay required for critical care patients.

To evaluate systemic fragility, a Monte Carlo risk engine executed 100 stochastic surge arrival iterations. Under baseline capacities (239 Standard Beds, 31 ICU Beds), the Monte Carlo evaluation revealed a **100.0% probability of ICU capacity failure**, proving that unmitigated emergency surges inevitably lead to critical care exhaustion.

6.4 User Interface & Prescriptive Analytics Integration

The user interface was compiled into a single-page interactive application ([dashboard.py](#)) using Streamlit, Pandas, and Matplotlib. The interface connects user inputs directly to the simulation engine, instantly re-calculating active capacities, identifying hourly breach points, displaying financial metrics, and triggering prescriptive cost-saving guidance.

Section 7: Evaluation & Scenario Analysis

To test system efficacy and demonstrate decision support capabilities, four operational scenarios were simulated and evaluated within the Streamlit dashboard environment.

7.1 Quantitative Scenario Comparison Matrix

Metric / Parameter	Scenario 1: Baseline (Status Quo)	Scenario 2: Break-Even Conversion	Scenario 3: Fully Optimised Buffer (Recommended)	Scenario 4: External Rental Alternative
Rented Standard Beds (\$500/ea)	0	0	0	0
Rented ICU Units (\$2,000/ea)	0	0	0	50
Converted Standard Beds (\$800/ea)	0	44	50	0
Active Standard Care Capacity	239 beds	195 beds	189 beds	239 beds
Active ICU Bed Capacity	31 beds	75 beds	81 beds	81 beds
Peak Standard Utilisation	172 beds	172 beds	172 beds	172 beds
Peak ICU Utilisation	75 beds	75 beds	75 beds	75 beds
Idle Standard Buffer at Peak	67 beds	23 beds	17 beds	67 beds
ICU Headroom / Safety Buffer	-44 beds (Deficit)	0 beds	+6 beds	+6 beds
Hourly Capacity Breach Periods	20 Hours	0 Hours	0 Hours	0 Hours
Total Operational Cost	\$0	\$35,200	\$40,000	\$100,000
System Evaluation Status	SYSTEM FAILURE	SYSTEM OPTIMIZED	SYSTEM OPTIMIZED	SYSTEM OPTIMIZED

7.2 Detailed Scenario Breakdown & Dashboard Visuals

Scenario 1: Baseline Unmitigated Surge (Status Quo)

- **Operational Parameters:** External Standard Rentals = 0; External ICU Rentals = 0; Converted Beds = 0.
- **Active Capacities:** Active Standard Care Capacity = 239 beds; Active ICU Capacity = 31 beds.
- **Operational Findings:** Standard care floor utilisation remains safe throughout the surge, peaking at 172 beds occupied and leaving 67 unutilised standard beds at peak surge. However, ICU bed demand breaches the 31-bed baseline capacity at Hour 4 (33 occupied beds) and escalates to a peak of 75 occupied beds at Hour 20 (a 44-bed critical deficit).
- **System Evaluation:** System status triggers **SYSTEM FAILURE** due to capacity breaches across **20 out of 24-hourly periods**.



Figure 4A: Baseline UI Dashboard — Prescriptive Suggestion and Capacity KPI Metrics

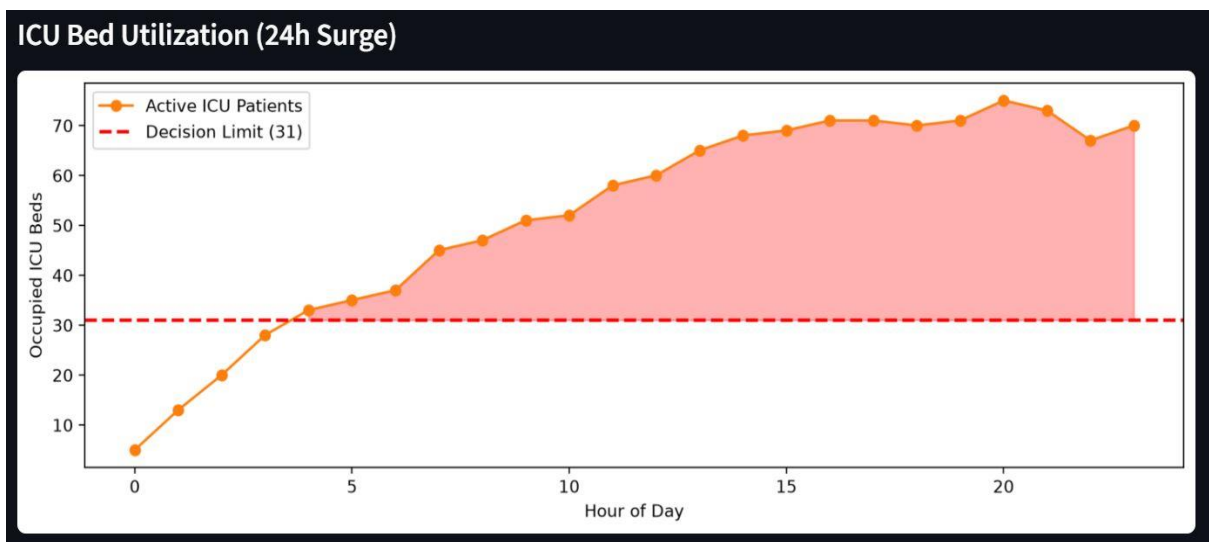


Figure 4B: Baseline ICU Capacity Utilisation — Hourly Red-Zone Breach Fill

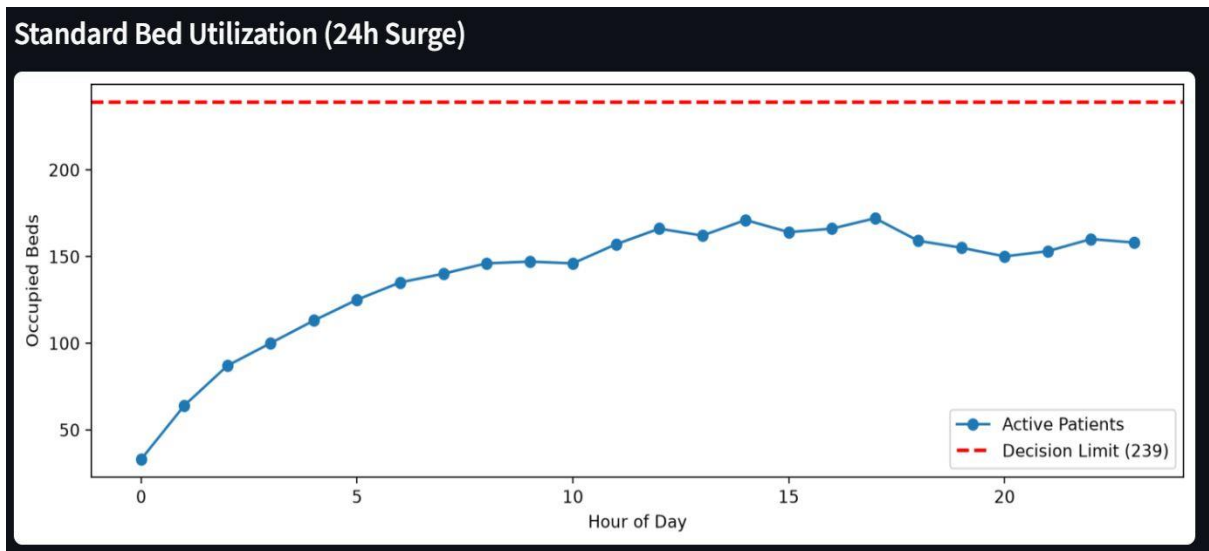


Figure 4C: Baseline Standard Floor Utilisation Curve

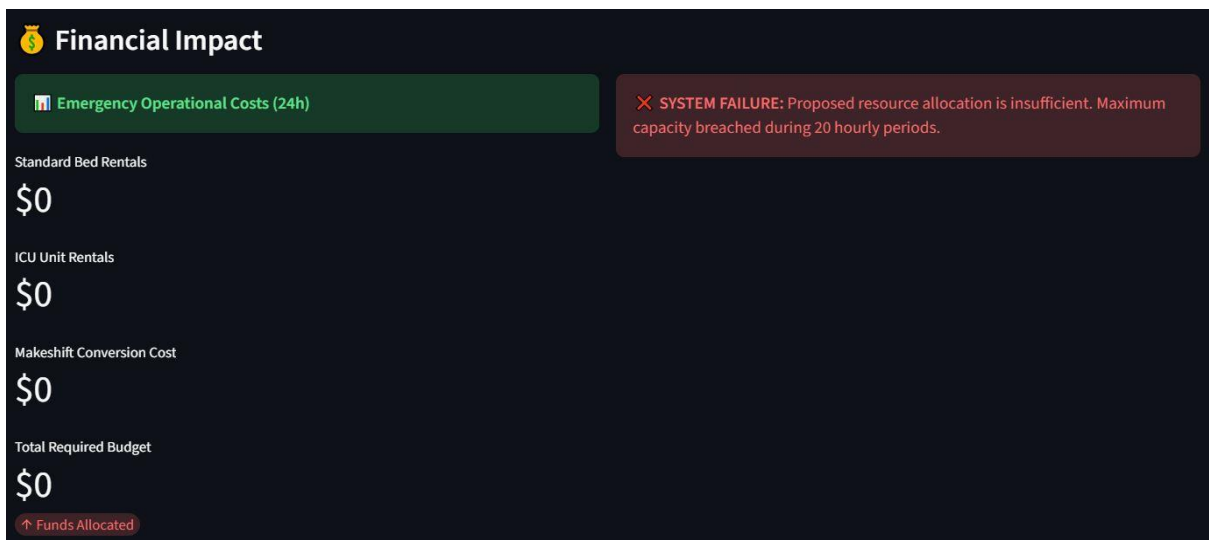


Figure 4D: Baseline Financial Output & System Failure Alert Banner

Scenario 2: Break-Even Conversion Strategy

- **Operational Parameters:** Convert 44 Idle Standard Beds (\$800/ea); External Rentals = 0.
- **Active Capacities:** Active Standard Care Capacity = 195 beds ($239 - 44$); Active ICU Capacity = 75 beds ($31 + 44$); Idle Standard Buffer = 23 beds ($67 - 44$).
- **Operational Findings:** Reallocating 44 idle standard care beds reduces standard floor capacity to 195 beds, which safely absorbs peak standard floor demand (172 beds). ICU capacity expands to 75 beds, perfectly matching peak ICU demand at Hour 20.
- **System Evaluation:** All 20 breach periods are completely eliminated at a total operational cost of **\$35,200** ($44 \times \800).



Figure 5A: Break-Even UI Dashboard — Reallocated Capacity Indicators (44 Beds Converted)

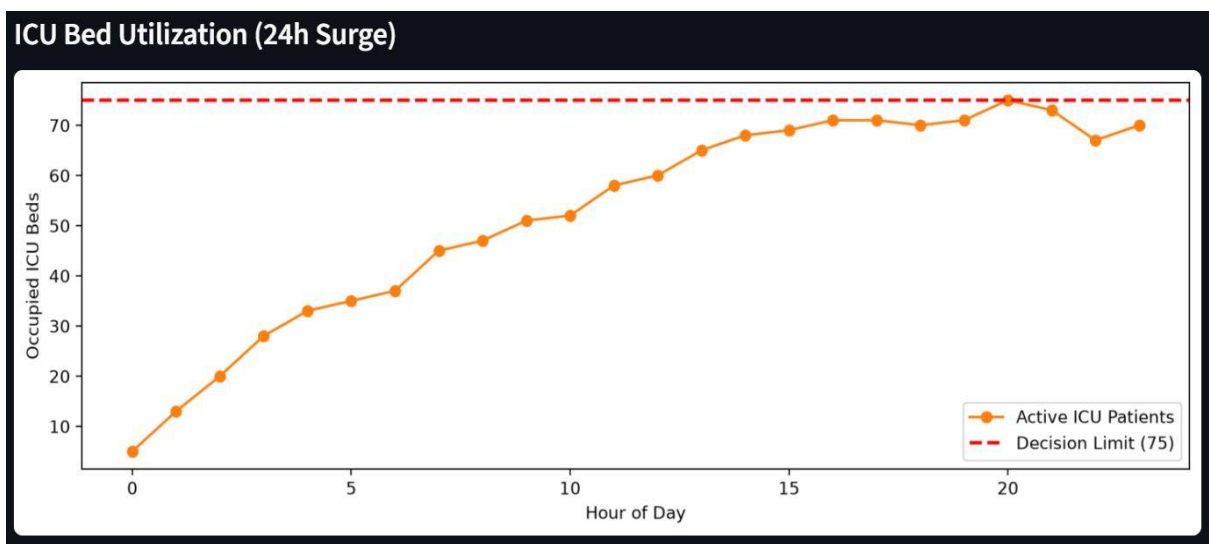


Figure 5B: Break-Even ICU Bed Utilisation Curve — Breach Eliminated

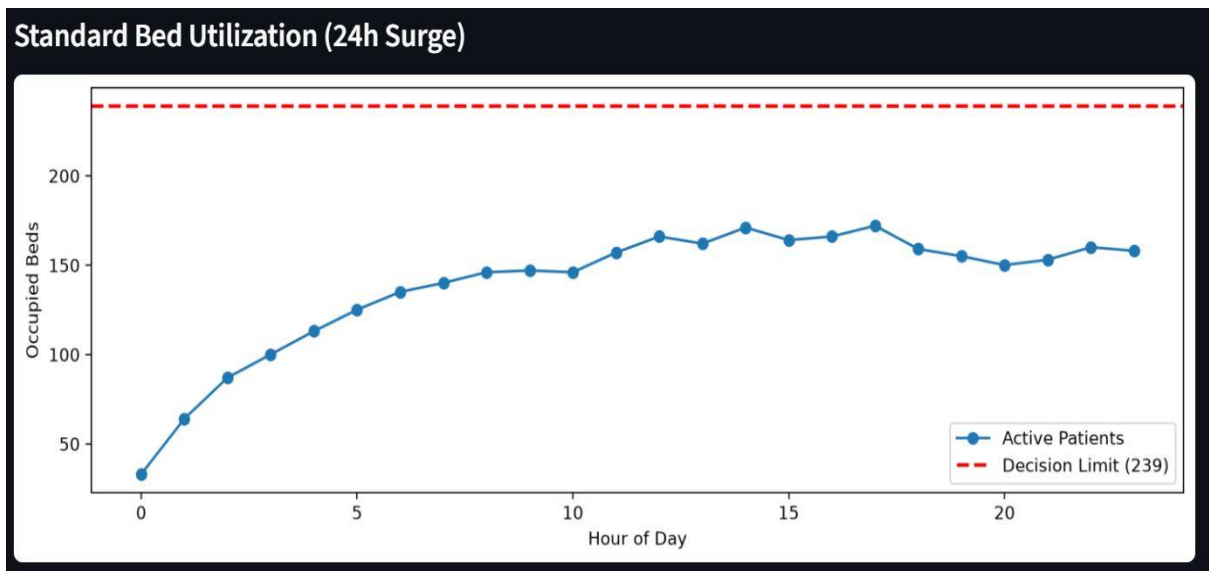


Figure 5C: Adjusted Standard Floor Capacity Curve

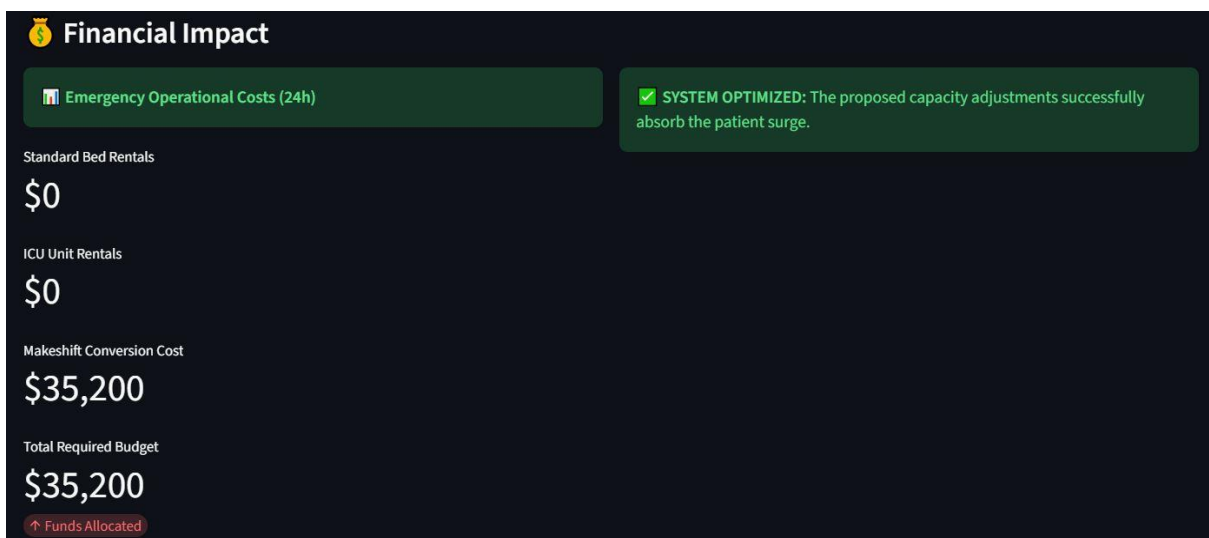


Figure 5D: Break-Even Financial Impact (\$35,200 Operational Expenditure)

Scenario 3: Fully Optimised Buffer Strategy (Recommended Solution)

- **Operational Parameters:** Convert 50 Idle Standard Beds (\$800/ea); External Rentals = 0.
- **Active Capacities:** Active Standard Care Capacity = 189 beds (239 – 50); Active ICU Capacity = 81 beds (31 + 50); Idle Standard Buffer = 17 beds (67 – 50).
- **Operational Findings:** Converting 50 standard floor beds expands active ICU capacity to 81 beds, establishing a **6-bed safety headroom buffer** above peak ICU demand (75 beds). Standard care floor space retains 17 unutilised buffer beds above its peak demand (172 beds).
- **System Evaluation:** System status displays **SYSTEM OPTIMIZED**. The total expenditure of **\$40,000** provides robust operational protection against unexpected secondary arrival spikes during the surge window.



Figure 6A: Recommended Strategy UI Dashboard — 50 Converted Standard Beds with 17 Buffer Beds

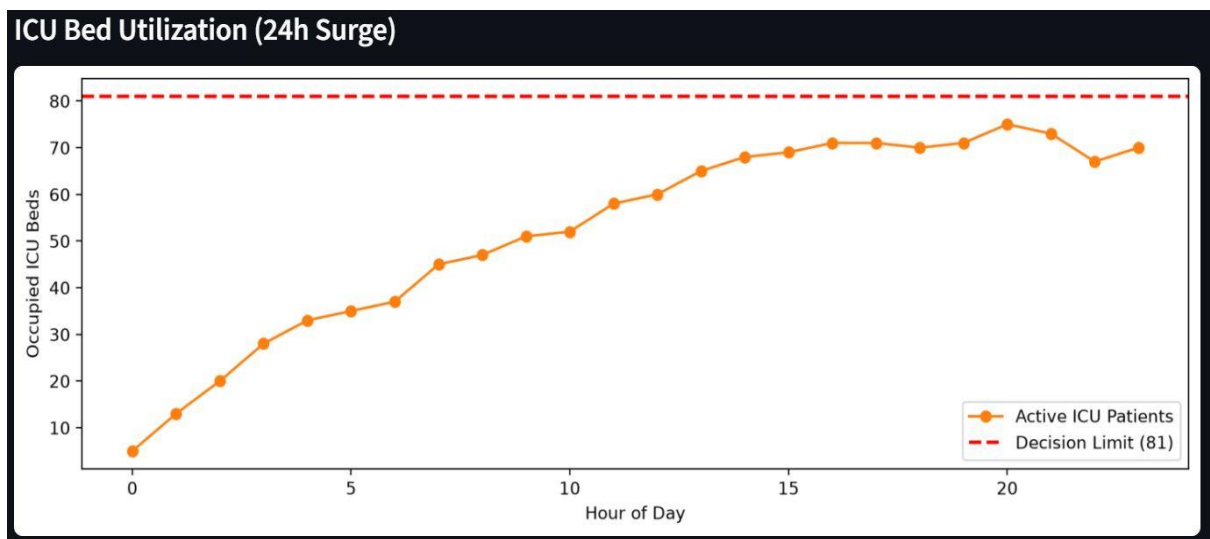


Figure 6B: Fully Optimised ICU Bed Utilisation — 81 Capacity Limit vs. 75 Peak Occupancy

Standard Bed Utilization (24h Surge)

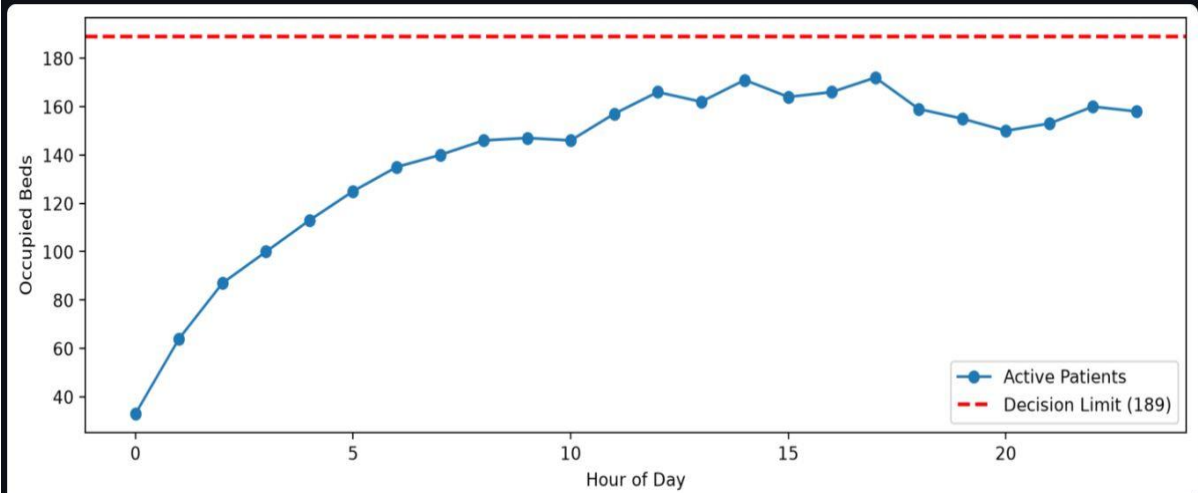


Figure 6C: Optimised Standard Floor Bed Utilisation Curve

Financial Impact

Emergency Operational Costs (24h)

✓ **SYSTEM OPTIMIZED:** The proposed capacity adjustments successfully absorb the patient surge.

Standard Bed Rentals

\$0

ICU Unit Rentals

\$0

Makeshift Conversion Cost

\$40,000

Total Required Budget

\$40,000

↑ Funds Allocated

Figure 6D: Optimised Strategy Financial Breakdown (\$40,000 Budget Allocation)

Scenario 4: Un-optimised External Rental Alternative

- **Operational Parameters:** Rent 50 External ICU Units (\$2,000/ea); Converted Beds = 0.
- **Active Capacities:** Active Standard Care Capacity = 239 beds; Active ICU Capacity = 81 beds (31 + 50); Idle Standard Buffer = 67 beds.
- **Operational Findings:** External rentals achieve an identical 6-bed ICU safety buffer (81 capacity vs. 75 peak occupied). However, leaving 67 standard care beds completely idle while procuring expensive external rental units represents severe financial inefficiency.
- **System Evaluation:** Operational expenditure rises to **\$100,000** (50 × \$2,000). Comparative DSS analysis demonstrates that internal asset conversion (Scenario 3) delivers identical clinical safety while saving **\$60,000 (60.0% cost reduction)**.



Figure 7A: External Rental Strategy UI Dashboard — 50 External ICU Units Procured

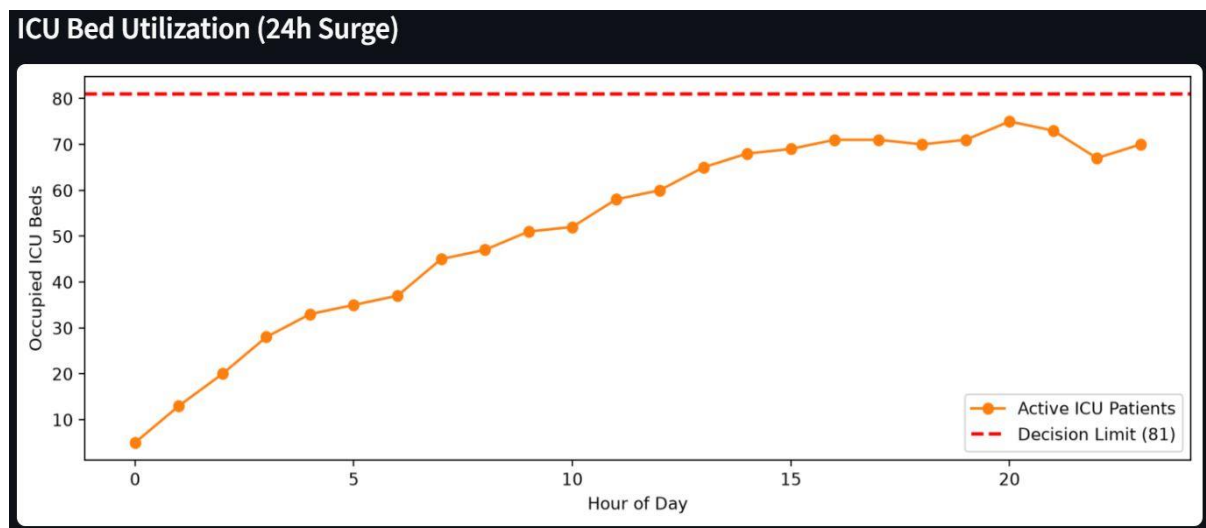


Figure 7B: External Rental ICU Bed Utilisation Curve

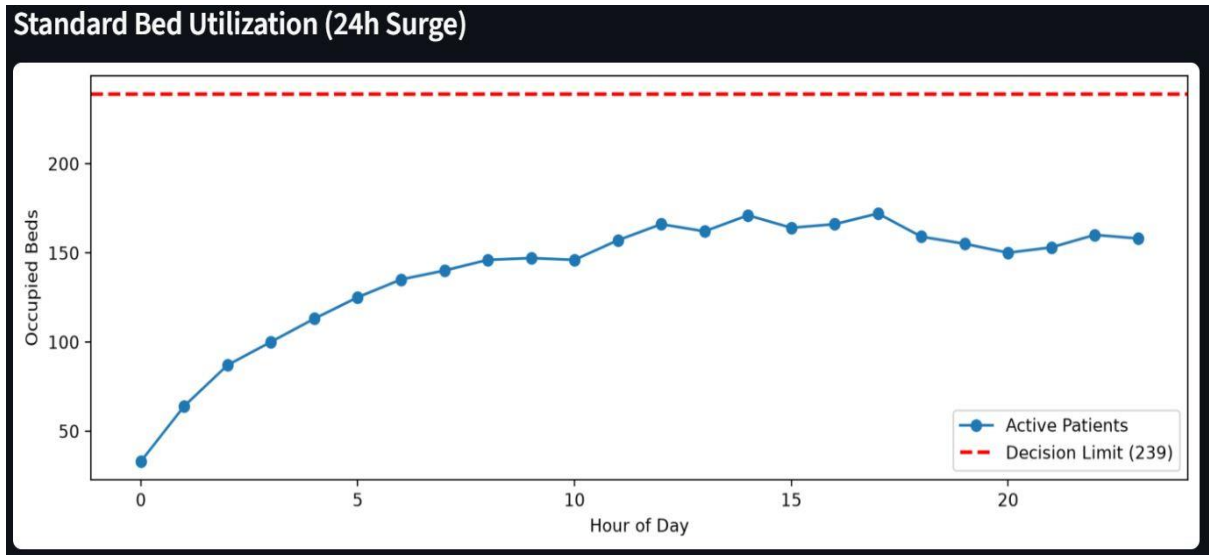


Figure 7C: Standard Floor Bed Utilisation Curve with High Idle Capacity

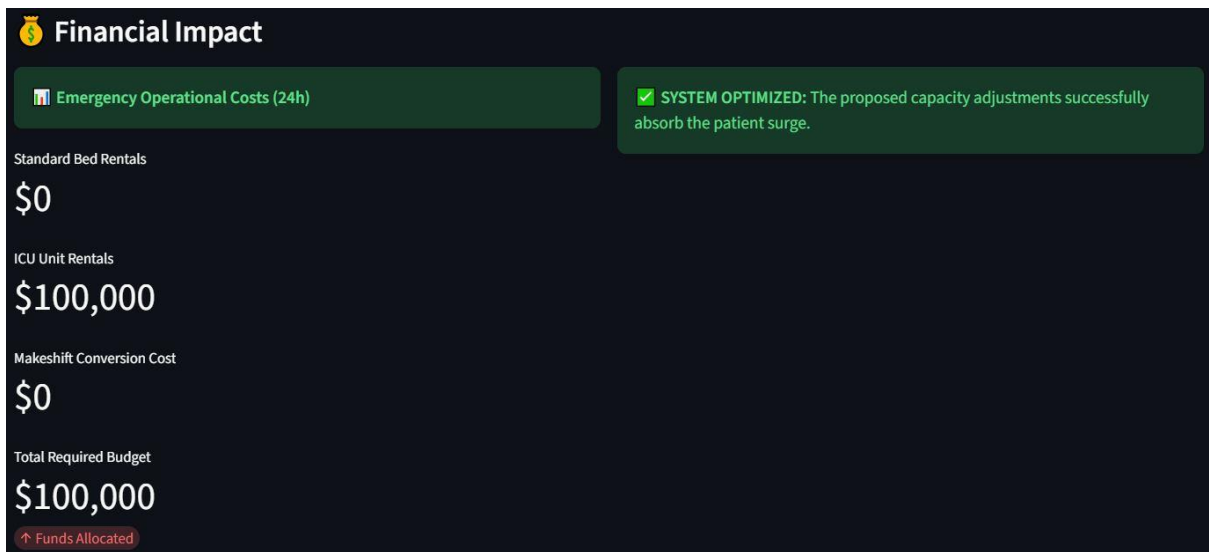


Figure 7D: External Rental Financial Impact (\$100,000 Operational Expenditure)

Section 8: Critical Discussion

8.1 System Strengths

- **Seamless Data Integration:** Successfully unifies cloud PostgreSQL database tables with international medical coding crosswalks (ICD-10 to ICD-11) and live weather API data.
- **Practical Hybrid Analytics:** Combining Random Forest machine learning with queuing logic captures real-world surge trends far more accurately than static spreadsheets.
- **Clear Financial Guidance:** Rather than simply sounding an alarm when beds run out, the system actively identifies unused ward space and recommends lower-cost bed conversions (\$800 vs \$2,000).

8.2 System Limitations

- **Instant Conversion Assumption:** The current mathematical model assumes converting a standard bed to an ICU bed happens instantly at \$800. In practice, physical room setup, ventilator positioning, and specialist nurse reassignments take time.
- **Staffing Constraints:** The simulation focuses on physical beds and equipment but does not yet enforce mandatory nurse-to-patient ratios (e.g., 1:1 or 1:2 ICU nurse coverage).

8.3 Decision-Quality Improvements over Traditional Methods

- **Proactive Horizon vs. Reactive Bed Management:** Traditional bed management reacts after emergency department boarding queues overflow. This DSS provides a 24-hour predictive planning horizon, enabling proactive resource reconfiguration prior to peak arrivals.
- **Elimination of Capital Waste:** By identifying unutilised standard care floor space, the DSS prevents unnecessary external equipment rentals, saving \$60,000 per 24-hour surge event.
- **Inter-Departmental Administrative Consensus:** Connects clinical metrics (CMO focus on zero boarding delays) with operational expenditure (CFO focus on budget control) within a single unified visual interface.

Section 9: Ethical & Organisational Considerations

9.1 Fairness and Algorithmic Bias Mitigation

- **Triage Protocol Integrity:** Algorithmic forecasting models used in clinical environments must not introduce socioeconomic, demographic, or origin-based biases into medical triage processes. The Random Forest regressor engineered within this DSS relies strictly on non-identifiable temporal, clinical resource type, and meteorological variables (e.g., historical temperature, day of week, region origin tag) to maintain absolute triage equity.
- **Cross-Regional Standardised Care:** The relational translation crosswalk enforces uniform resource assignment and triage logic across all origin regional hubs (US, EU, Dubai), preventing localised operational discrimination or unequal patient prioritisation.

9.2 Transparency and Model Explainability

- **Glass-Box Simulation Architecture:** Healthcare leadership and clinical staff often reject "black-box" decision tools. To ensure complete operational transparency, the DSS utilises glass-box queuing logic and explicit, rule-based cost equations. The UI displays clear time-series occupancy curves alongside active decision limits, allowing hospital administrators to verify exactly how breach warnings and financial recommendations are derived.

9.3 Accountability and Human-in-the-Loop Governance

- **Prescriptive Advisory Role:** The DSS operates strictly as a prescriptive decision-support platform rather than an automated decision-maker. Final operational authorisation for emergency budget allocation, physical room reconfiguration, and equipment procurement remains firmly with human hospital executive leadership.

9.4 Organisational Acceptance and System Adoption

- **Role-Based Operational Training:** Successful deployment requires cross-functional operational training across bed logistics managers, nursing supervisors, and finance teams to align workflows around predictive surge insights.
- **Building Executive Trust:** Demonstrating quantifiable cost savings (\$60,000 cost avoidance per surge event) alongside clinical risk reduction establishes strong organisational confidence and accelerates administrative adoption.

Section 10: Conclusion & Recommendations

10.1 Summary of Findings

1. **Unmitigated Baseline Vulnerability:** Under baseline operations (239 Standard Care Beds, 31 ICU Beds), emergency patient surges result in severe ICU capacity failure across 20 out of 24-hourly periods, reaching a peak deficit of 44 ICU beds at Hour 20.
2. **Cost-Effectiveness of Internal Asset Conversion:** Converting idle standard care floor space into temporary makeshift ICU units (\$800/bed) resolves 100% of capacity breaches at a fraction of the cost of external equipment procurement.
3. **Recommended Operational Strategy (Scenario 3):** Converting **50 idle standard beds** establishes an active operational capacity of 189 Standard Care Beds and 81 ICU Beds. This absorbs peak surge demand while maintaining a 6-bed ICU safety buffer and a 17-bed standard floor buffer for a total budget allocation of **\$40,000**.
4. **Demonstrated Value:** Compared to an un-optimised external rental alternative (\$100,000), the DSS-guided conversion strategy achieves a **\$60,000 (60.0%) operational cost reduction** per 24-hour surge event.

10.2 Strategic Recommendations for Executive Leadership

1. **Adopt Prescriptive Asset Conversion Protocols:** Implement administrative policies authorising bed logistics managers to convert idle standard floor space into critical care stations whenever predictive models forecast an upcoming ICU capacity deficit.
2. **Establish Pre-Approved Emergency Contingency Funds:** Allocate a pre-approved \$40,000 emergency response fund per anticipated surge event to enable rapid floor conversions without bureaucratic procurement delays.
3. **Integrate Real-Time Bed Sensor Telemetry:** Connect automated hospital bed telemetry and electronic health record (EHR) discharge logs directly into the Neon PostgreSQL cloud database pipeline for continuous real-time occupancy tracking.

10.3 Future Technical Enhancements

- **Clinical Staffing Queue Integration:** Incorporate clinical nurse shift schedules (`hospital_staff`, `staff_schedule`) into the discrete-event queue simulation to enforce mandatory nurse-to-patient staffing ratio constraints.
- **Dynamic Supply-Chain Procurement API Feeds:** Connect live supply-chain APIs to update equipment rental and conversion costs dynamically based on real-time market rates.
- **Mathematical Optimisation Solver Module:** Integrate an Integer Linear Programming (ILP) optimisation module to automatically recommend specific floor-by-floor spatial conversions across individual hospital wings.